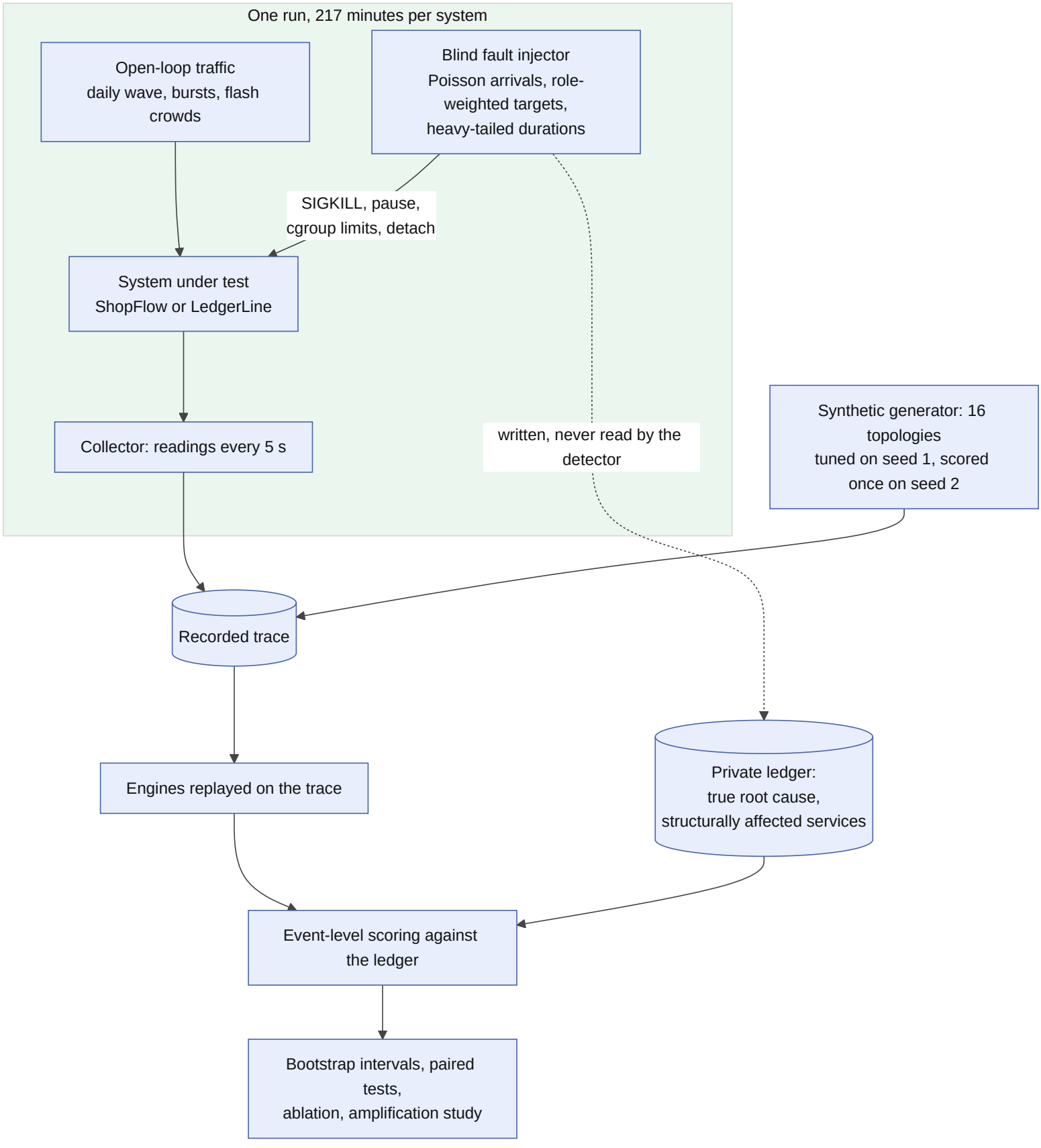
One run, 217 minutes per system
Open-loop traffic
daily wave, bursts, flash crowds
Blind fault injector
Poisson arrivals, role-weighted targets, heavy-tailed durations
SIGKILL, pause, cgroup limits, detach
System under test
ShopFlow or LedgerLine
Collector: readings every 5 s
written, never read by the detector
Synthetic generator: 16 topologies
tuned on seed 1, scored once on seed 2
Recorded trace
Engines replayed on the trace
Private ledger:
true root cause,
structurally affected services
Event-level scoring against the ledger
Bootstrap intervals, paired tests,
ablation, amplification study